Software Engg. MSE 2 Solutions — 15/09/2025. Divya Surve.

VIT Vidyalankar Institute of Technology
(Accredited A+ by NAAC)
(Autonomous Institute Affiliated to University of Mumbai)

Mid Semester Examination

Branch	Date	Sem.	Roll No. / Exam Seat No.	Subject	Student's Signature	Junior Supervisor's Name and Sign

Question No.	A	B	C	D	E	F	G	H	Total	Total out of (20 /30 / 40)
1										
2										
3										
4										

Examiners Signature	Student's Sign (After receiving the assessed answer sheet)

Q1 a) Hospital Management System Development has modules.
- Patient Registration
- Billing
- Report Generation.

Tasks for each module
- Patient Registration
  - T1 - Requirement Analysis
  - T2 - Database Design
  - T3 - UI Development.
  - T4 - Integration & Testing
- Billing
  - T5 - Requirement Analysis
  - T6 - Database Design
  - T7 - UI Development
  - T8 - Integration & Testing.
- Report Generation
  - T9 - Requirement Analysis
  - T10 - Database Design
  - T11 - UI Development
  - T12 - Integration Testing.

Task sets :-

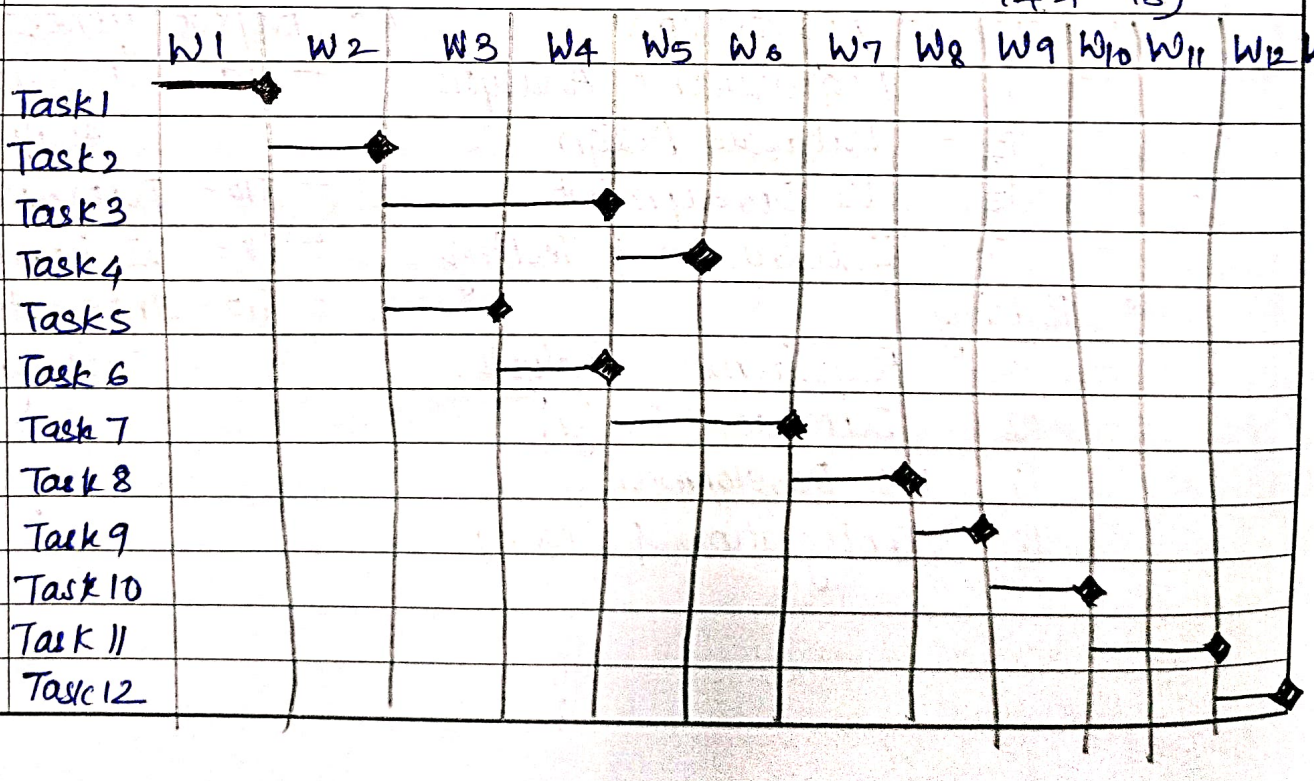
Each module's task set is collection of its activities
- Patient Registration Task set $\{T_1, T_2, T_3, T_4\}$
- Billing Task set $\{T_5, T_6, T_7, T_8\}$
- Report Generation Task set $\{T_9, T_{10}, T_{11}, T_{12}\}$

Task Network :-

$T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow T_4$ (Patient Registration flow)
T_5 depends on completion of T_2 (Patient DB) because billing uses patient data.
T_9 depends on T_4 & T_8 (reports need both registration & billing).

The task network looks like :-

Patient Registration : $T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow T_4$
Billing : $T_5 \rightarrow T_6 \rightarrow T_7 \rightarrow T_8$ (T_5 depends on T_2)
Report Generation : $T_9 \rightarrow T_{10} \rightarrow T_{11} \rightarrow T_{12}$ (T_9 depends on T_4 & T_8)



Schedule Tracking : At the end of each week check
- planned finish & actual finish week of tasks.
- If T3 slips & end in W6 instead of W5 → T4 is delayed - Billing & Report module may also shift.
- It helps updating dependencies & revising deadlines early.

Resource Allocation :-

Resource	Assigned Task
Analyst	T1, T5, T9
DB Engineer	T2, T6, T10
UI Developer	T3, T7, T11
Tester	T4, T8, T12
Project Manager	Schedule tracking, reviews

Q| B).

Risk Exposure is expected monetary value of a risk.

$RE = P \times C$

where P- probability of risk occurance
C- cost of risk occur.

Calculate RE for each risk.

Risk A. (Payment Gateway)

$RE_A = 0.25 \times 5,00,000 = 1,25,000/-$

Risk B (Shopping cart defects)

$RE_B = 0.30 \times 3,50,000 = 1,05,000/-$

Risk C (Authentication legal issue)

$RE_C = 0.15 \times 8,00,000 = 1,20,000/-$

3. Ranking of Risks (Highest RE → Lowest RE)

1. Risk A — 1,25,000/-
2. Risk C — 1,20,000/-
3. Risk B — 1,05,000/-

4. Total Risk Exposure —

$RE_{Total} = RE_A + RE_B + RE_C$

= Rs 3,50,000/-

Q1. c)

Function Point Estimation	LOC Estimation
- Estimates cost based on functionality delivered to user	Estimates cost based on size of system measured in source code lines.
- Independent of programming language, tools or coding style.	- Highly dependent on programming like 1000 LOC in C may equal only 200 LOC in Python.
- It can be used early in software life cycle since it measures functions not code.	- It can be used after design / coding when the system's size in LOC can be reasonably predicted.
- More accurate for early estimation as it considers users requirement directly	- More accurate after implementation but poor for early prediction.

Converts function points into effort using productivity rates	converts LOC into effort using productivity rates
Supports productivity measurement across languages.	cannot compare productivity across different languages.

Q2 A).

Each week BCWS, at the end of 3 weeks cumulative BCWS = 30,000/- — Planned value.

BCWP at the end of 3 weeks - EV =
9000 + 8000 + 8000 = 25,000/-

Actual cost = 10,000 + 9,000 + 9,000 = 28,000/-

Total project budget = BAC = 10,000 X 5 = 50,000/-

SV = BCWP - BCWS = 25,000 - 30,000 = -5,000/-
behind schedule.

SPI = $\frac{25,000}{30,000}$ = 0.833 < 1 SPI < 1 — behind schedule.

CV = 25,000 - 28,000 = -3,000/- (negative → over budget)

CPI = $\frac{25000}{28000}$ = 0.89 < 1 (CPI < 1 → cost inefficient)

Percent completed = $\frac{25000}{50000}$ = 50%

Percent scheduled to be complete = $\frac{30000}{50000}$ x100 = 60%

The project is behind schedule
Planned to be 60% complete by week 3 but only 8% complete.

The project is over budget and cost inefficient so far.
Actual cost is greater than earned value

Q2 b)

LOC = 20,000$

Organic —

$$\text{Effort} = 2.4 \times (20)^{1.05}$$
$$= 55.75 \text{ person-month}$$

$$\text{Time} = 2.5 \ (55.75)^{0.38}$$
$$= 11.52 \text{ months}$$

$$\text{Persons} = \frac{55.75}{11.52} = \underline{4.83} \text{ persons}$$

Semi-detached-

$$\text{Effort} = 3.0 \times (20)^{1.12}$$
$$= 85.955 \text{ person-month}$$

$$\text{Time} = 2.5 \times (85.95)^{0.35}$$
$$= 11.88 \text{ months}.$$

$$\text{Persons} = \frac{85.95}{11.88} = \underline{7.25 \text{ persons}}$$

Embedded

$$\text{Effort} = 3.6 \times (20)^{1.20}$$
$$= 131.08 \text{ person-months.}$$

$$\text{Duration} = 2.5 (131.08)^{0.32}$$
$$= 11.900 \text{ months}$$

$$\text{Persons} = \frac{131.08}{11.900} = 11.01 \text{ persons.}$$

The effort grows with the type of project with the highest being for embedded project

The same applies to schedule too which increases slightly from 11.52 to 11.88 to 11.900 from basic to embedded project types.

Limitations of COCOMO model:-

Ignores cost driver attributes, does not consider factors like personnel capability, required reliability, product complexity. etc.

LOC is a problematic metric as its highly dependent on programming language, coding style and reuse. Hence, not suitable for non-procedural languages.